9.2 Two-Way Tables and Probability

Name _____ Period ____ Date _____

Algebra I 9.2 · Student sheet

I can…

- complete a two-way table.
- find joint and marginal relative frequencies.
- find conditional relative frequencies and use them to compare groups.

Vocabulary · fill in the missing word

Two-way table — a table that shows counts for two _____ at once

Joint frequency — a count _____ the table (one row and one column)

Marginal frequency — a total in the Total row or _____

Joint relative frequency — a joint frequency ÷ the _____ total

Marginal relative frequency — a marginal frequency ÷ the _____ total

Conditional relative frequency — a joint frequency ÷ the total of its row or _____

Part 1 · Class survey

Copy the class results and fill in the totals.

	Plays a sport	No sport	Total
Has a pet			
No pet			
Total			

a. What percent of the whole class has a pet and plays a sport? _____

b. What percent of the students who play a sport have a pet? _____

c. Why are the answers to a and b different? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Joint frequencies are _____ the table; **marginal frequencies** are the _____.

Example 1

A survey asked 80 students whether they own a pet. Complete the two-way table.

	Owns a pet	No pet	Total
9th grade	24	?	40
10th grade	?	12	?
Total	**52**	**?**	**80**

Relative frequencies: Joint relative frequency = cell ÷ _____ Marginal relative frequency = total ÷ _____

Example 2

Use the completed table.

a. What percent of the students are 10th graders who own a pet?

b. What percent of the students own a pet?

	Owns a pet	No pet	Total
9th grade	24	16	40
10th grade	28	12	40
Total	**52**	**28**	**80**

Conditional relative frequency: Conditional relative frequency = cell ÷ _____

"Given": The word "given" tells me which _____ to divide by.

Example 3

Use the completed table.

a. Given that a student is in 9th grade, what is the conditional relative frequency that the student owns a pet?

b. Given that a student is in 10th grade?

c. Given that a student owns a pet, what is the conditional relative frequency that the student is in 9th grade?

	Owns a pet	No pet	Total
9th grade	24	16	40
10th grade	28	12	40
Total	**52**	**28**	**80**

Part 3 · Partner practice

Partner 1

Complete the two-way table about a new lunch menu.

	Likes it	Doesn't	Total
Male	18	?	30
Female	?	8	?
Total	**40**	**?**	**?**

Partner 2

Use your completed table from Partner 1. Given that a student is female, what is the conditional relative frequency that she likes the menu? Given male?